

ESXi & vCenter Virtualisation Portfolio

Professional implementation and evidence documentation

Platform	VMware vSphere / ESXi
Management	VMware vCenter Server Appliance
Core services	Active Directory / DNS, NFS shared storage
Networking	vSwitches, port groups, VLANs and VMkernel networking
Availability	VM migration, DRS and vSphere HA
Evidence	21 implementation screenshots supplied with the project evidence

Prepared for professional portfolio / practical evidence

Mohammad Arif

Source: supplied ESXi/vCenter practical assessment and embedded evidence screenshots.

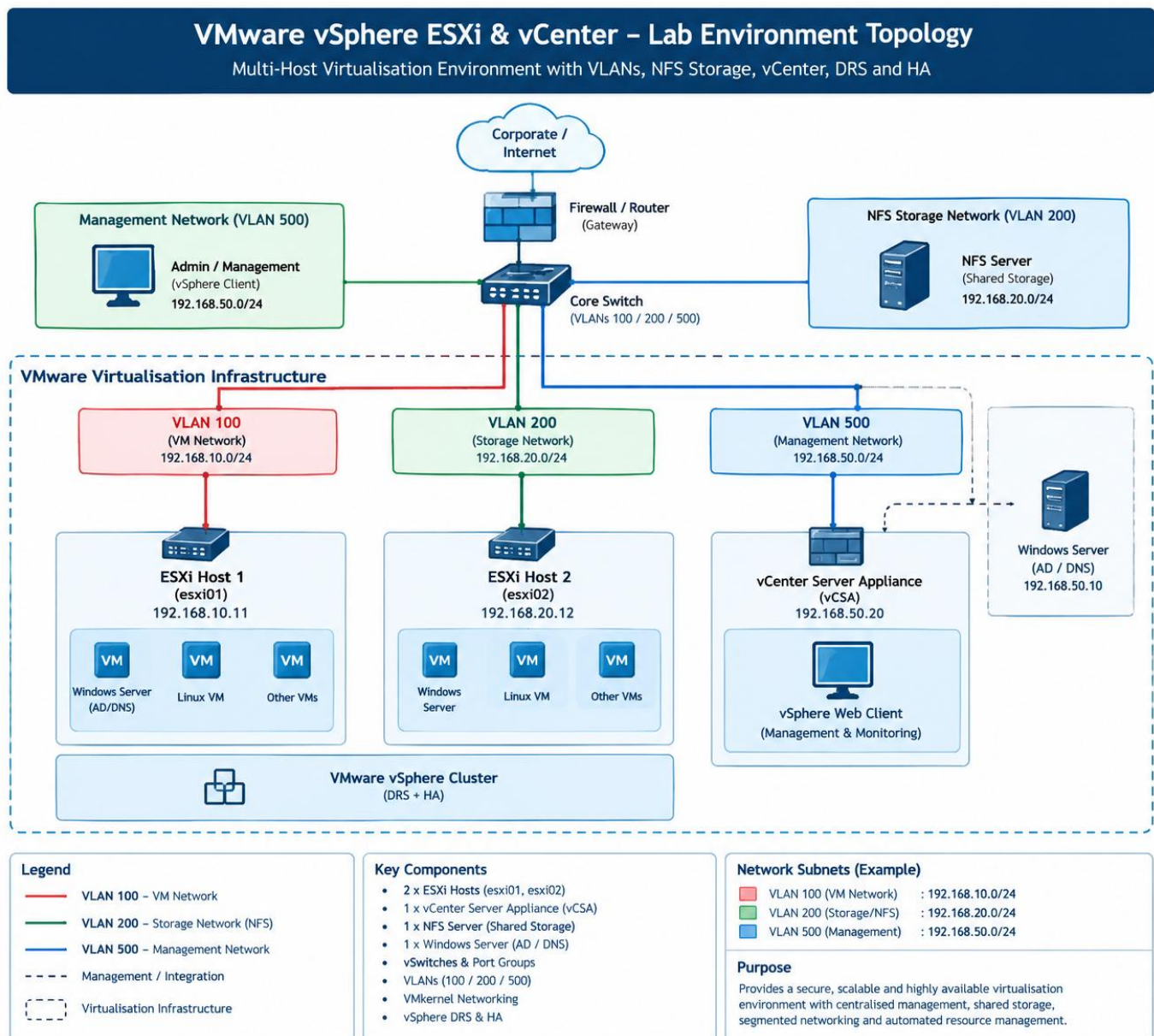
1. Project Overview

This portfolio documents the practical implementation of a private on-premise virtualisation environment using VMware vSphere ESXi and vCenter. The supplied project scenario required a multi-host ESXi environment, centralised vCenter management, virtual networking, shared NFS storage, virtual-machine provisioning, migration, DRS and High Availability.

The original assessment identifies the core implementation areas as installation of ESXi hosts and vCenter; creation of a virtual datacentre and cluster; VLANs, port groups and VMkernel networking; shared NFS storage; templates/clones; VM migration; and DRS/HA.

For portfolio use, each evidence screenshot is positioned immediately after the configuration activity it demonstrates. This makes the document suitable as a technical project record rather than a question-and-answer assessment submission.

Topology



2. Implementation Scope

- VMware ESXi hosts configured and managed as part of a vSphere environment.
- vCenter Server Appliance deployed for centralised management.
- Windows Server VM providing Active Directory and DNS services.
- NFS-based shared storage presented to the virtualisation environment.
- Virtual switches, port groups and VMkernel/network connectivity configured.
- Connectivity tested using Windows command-line ping and nslookup evidence.
- Virtual-machine provisioning, migration, DRS and vSphere HA demonstrated.

3. Environment at a Glance

The following details are transcribed from the supplied evidence screenshots and project brief.

Component	Evidence-visible detail
ESXi Host 1	esxi01.lab.local — 192.168.31.149 — ESXi 6.7.0 Update 3
ESXi Host 2	esxi2.lab.local — 192.168.31.129 — ESXi 6.7.0 Update 3
vCenter	vCenter Server Appliance 6.7.0.14000 — vcenter.lab.local — 192.168.31.20
DNS	dns.lab.local — 192.168.31.140
Virtual datacentre / cluster	LabDatacenter / New Cluster shown in vSphere Client evidence
Shared storage	NFS-Datastore / NFS server and share evidence
VLAN design from project brief	GPs VLAN 100; General Clinic Staff VLAN 200; IT Admins VLAN 500

4. High-Level Build Sequence

The supplied installation plan follows a staged approach: prepare hardware/networking; install ESXi; deploy vCenter; configure virtual networking; prepare shared storage; create and deploy VMs; enable HA/vMotion and backup; then test authentication, connectivity, performance, failover and restoration.

5. ESXi Hosts and vCenter

The first evidence group establishes the hypervisor hosts and central management layer. The screenshots show both ESXi hosts and the vCenter Server Appliance running as a VM.

5.1 ESXi Host 2

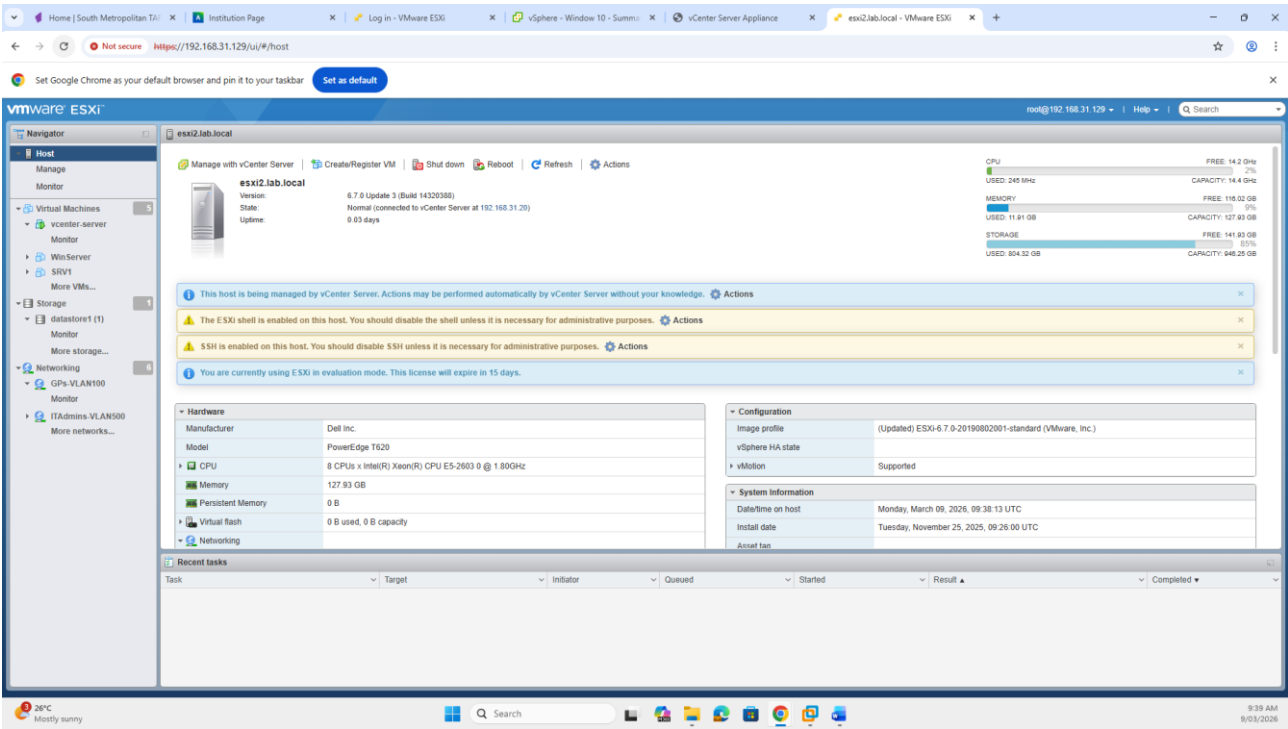


Figure 1 — ESXi host: esxi2.lab.local (192.168.31.129), ESXi 6.7.0 Update 3.

5.2 ESXi Host 1

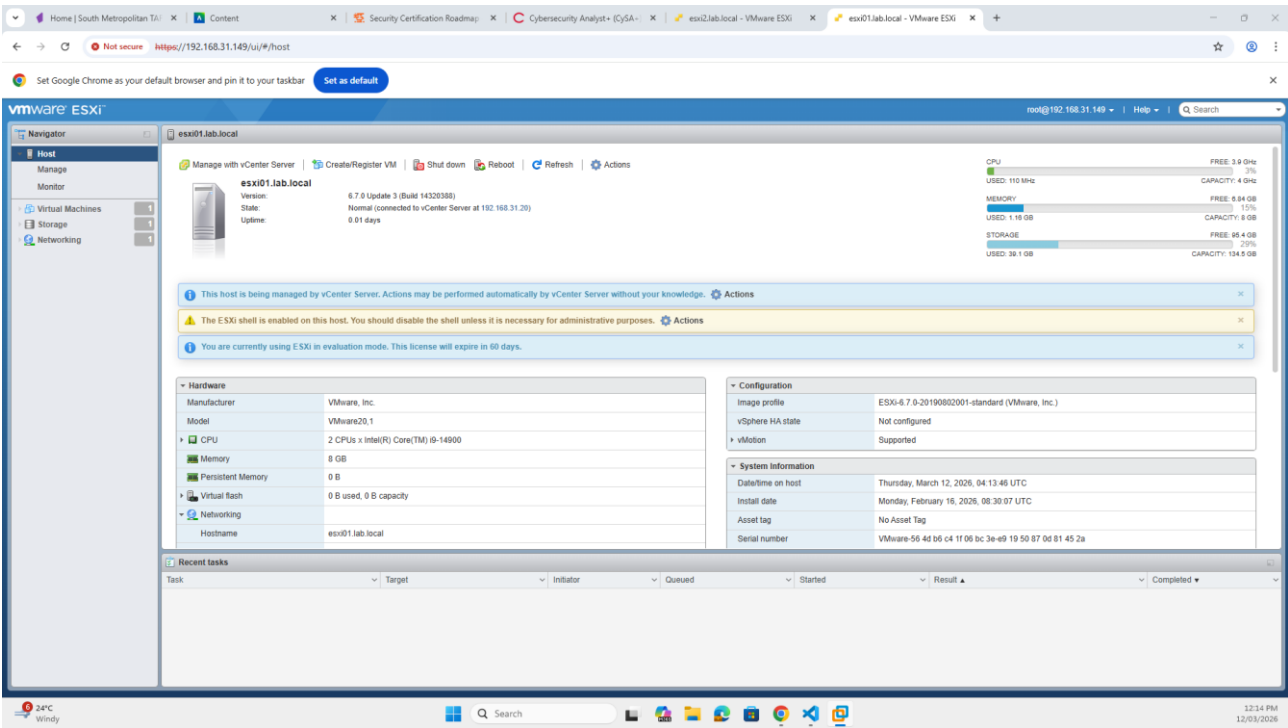


Figure 2 — ESXi host: esxi01.lab.local (192.168.31.149), ESXi 6.7.0 Update 3.

5.3 vCenter Server Appliance

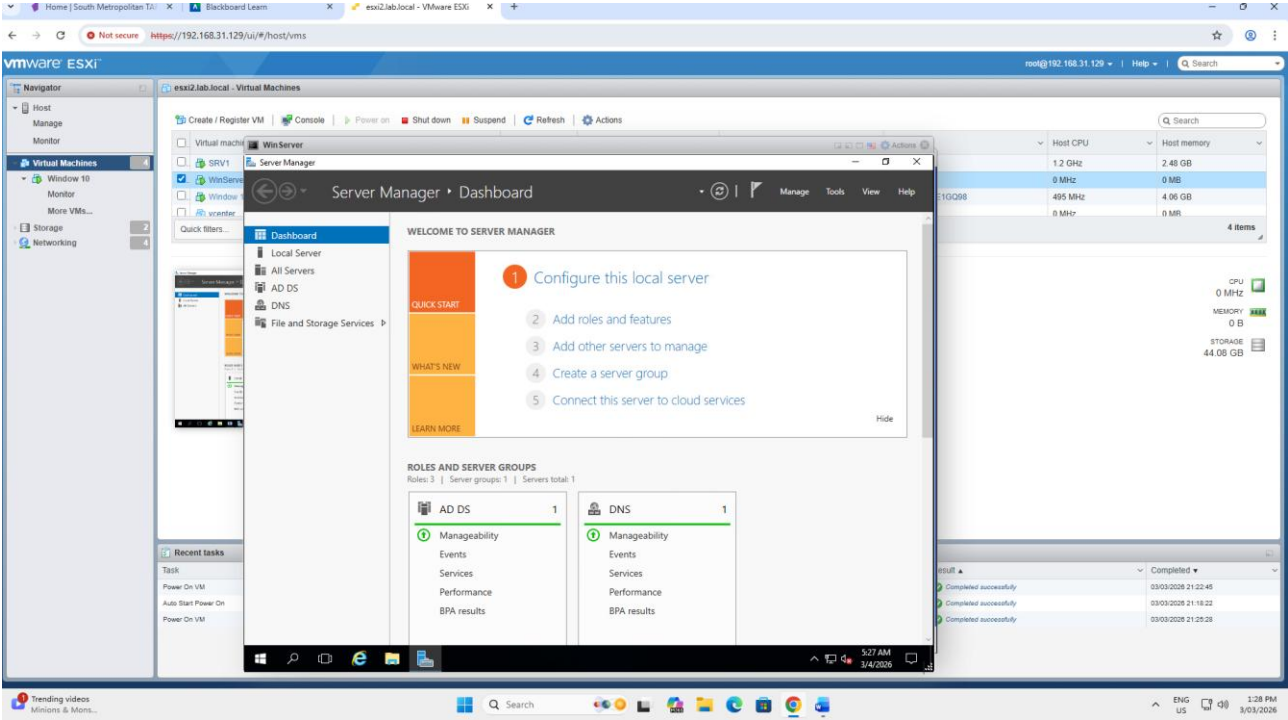


Figure 3 — vCenter Server Appliance console and VM hardware context.

5.4 Windows Server / Domain Services

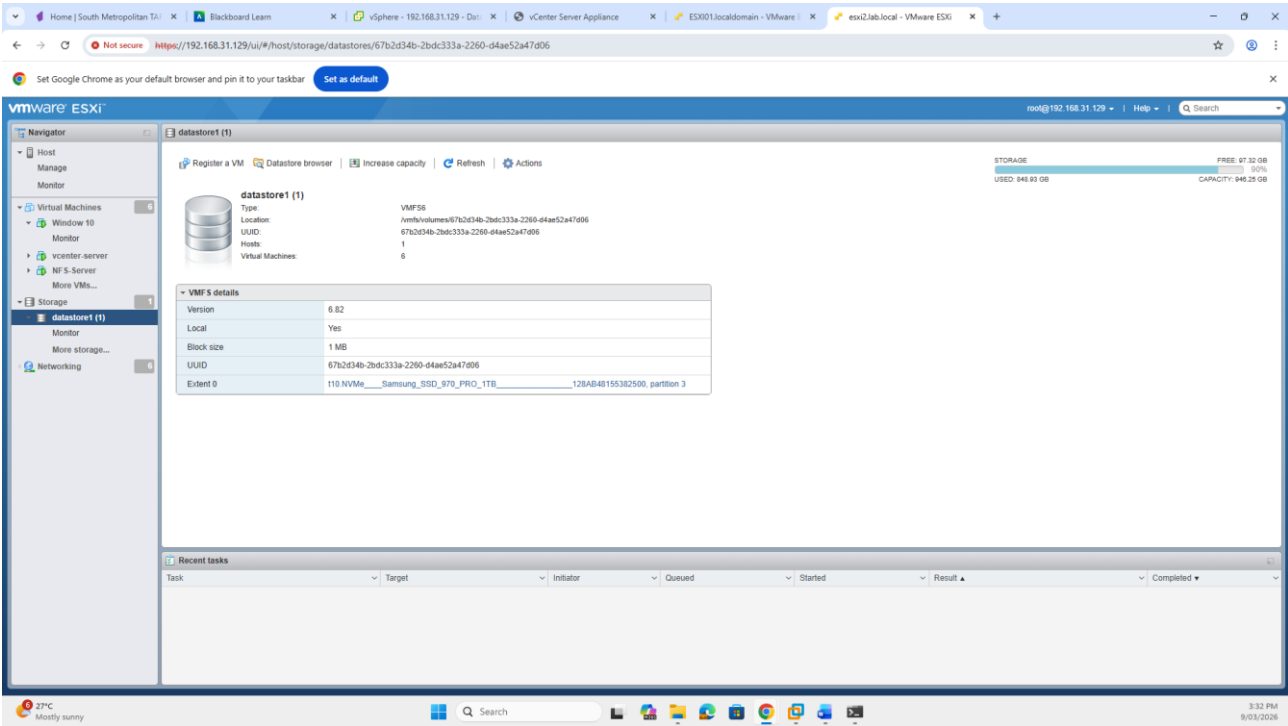


Figure 4 — Windows Server VM showing Active Directory Domain Services and DNS roles.

6. Virtual Datacentre, Cluster and Storage

The project requires a virtual datacentre and cluster for the ESXi hosts, together with shared storage for VM files and deployment resources. The supplied evidence includes vSphere inventory views and the NFS datastore/browser.

6.1 vSphere Inventory / Cluster Context

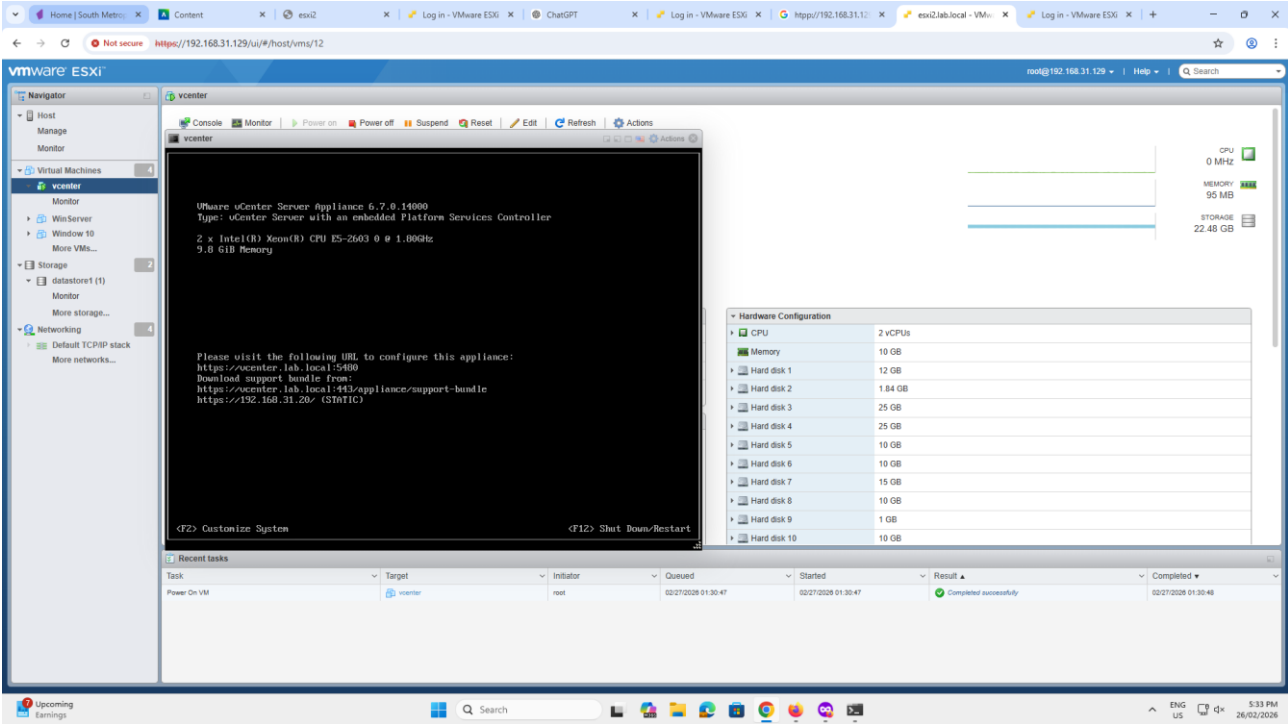


Figure 5 — vSphere Client inventory showing the managed virtualisation environment and cluster context.

6.2 Datastore Configuration

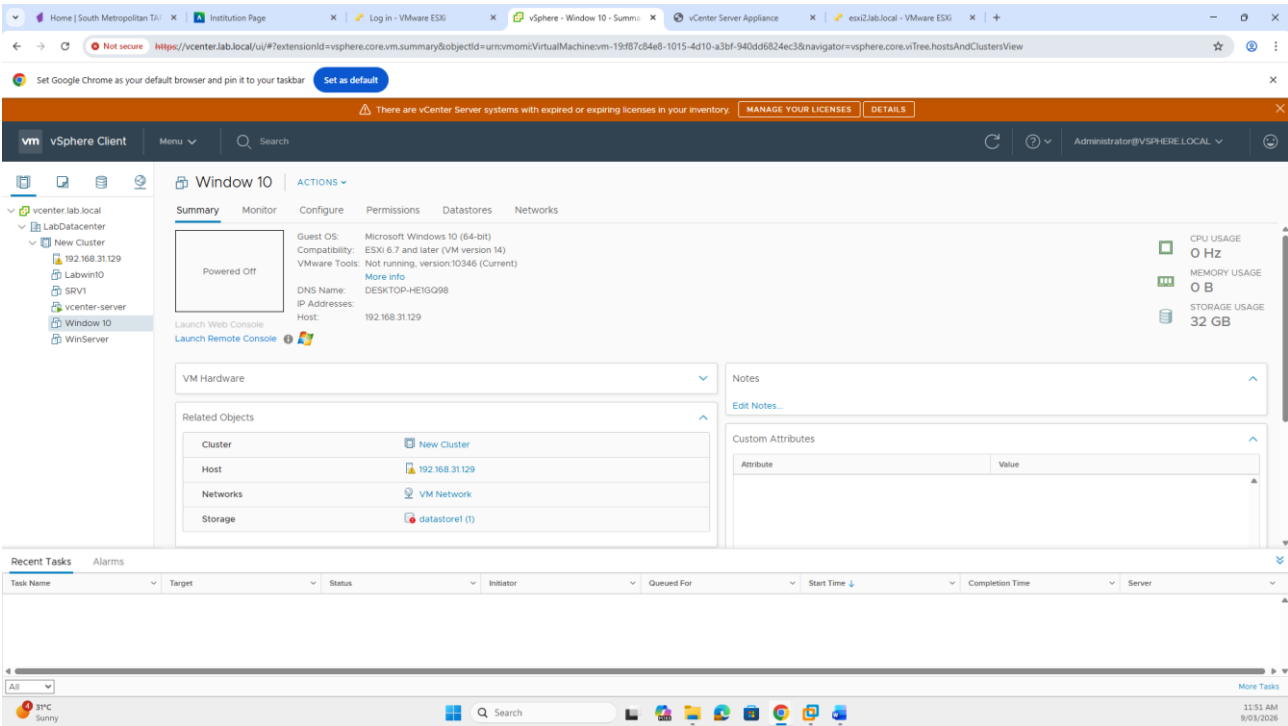


Figure 6 — ESXi storage view showing datastore configuration.

6.3 NFS Datastore Browser

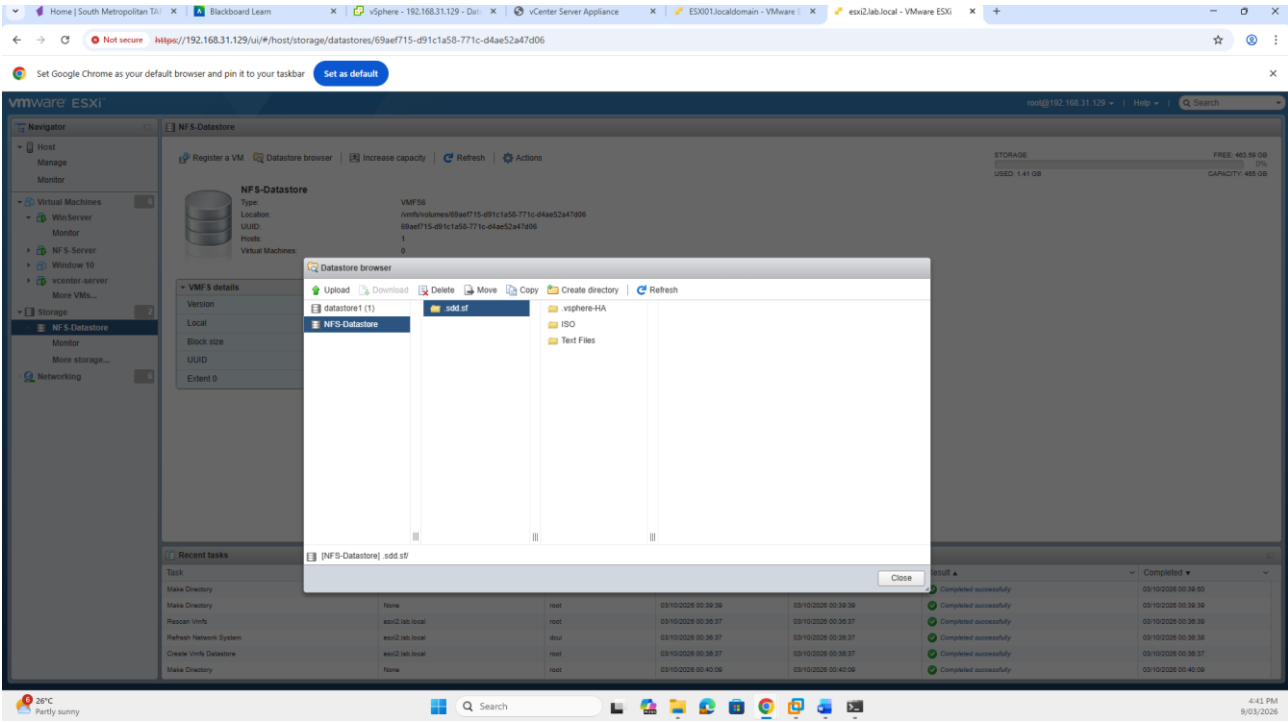


Figure 7 — NFS-Datastore browser showing shared datastore directories.

7. Virtual Networking

The networking implementation covers virtual switches, port groups, uplinks/VMkernel networking and connection of virtual machines to the required networks. The project also requires connectivity testing between virtual machines and from the management network to the classroom/internet environment.

7.1 Virtual Switches and Port Groups

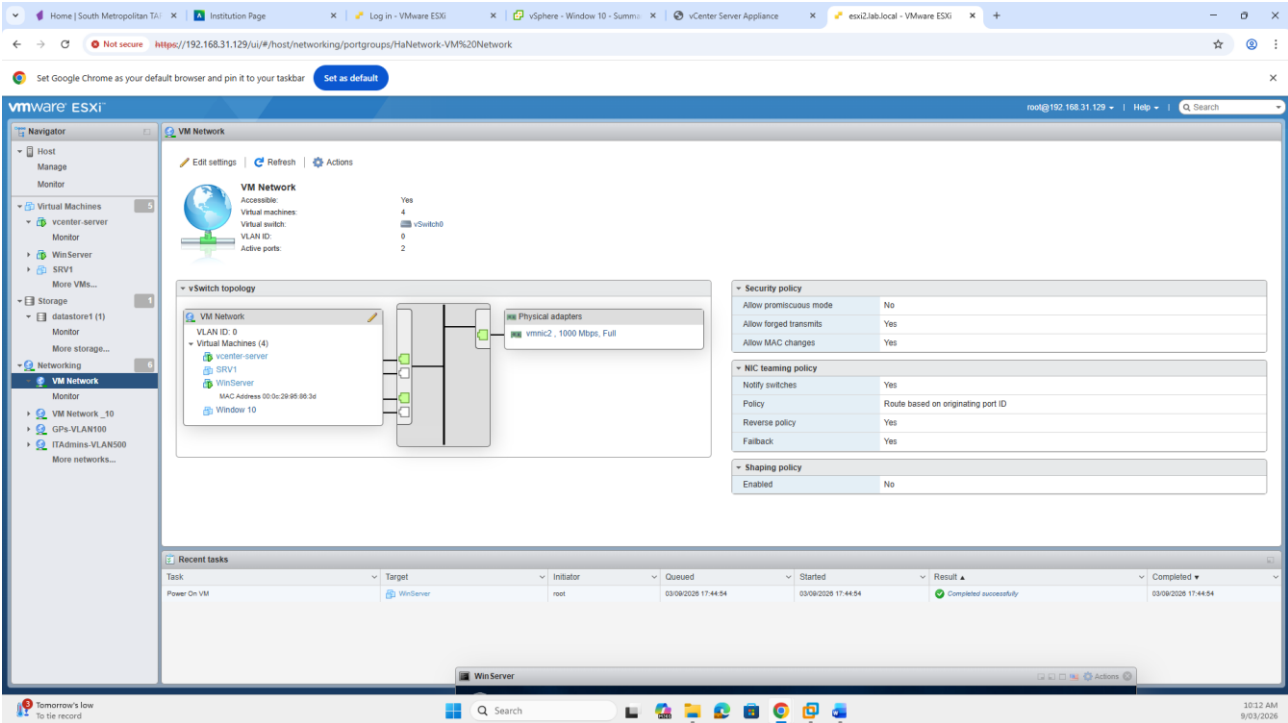


Figure 8 — ESXi networking view showing virtual switches, VMkernel NICs and port-group configuration.

7.2 VM Network / Port-Group Configuration

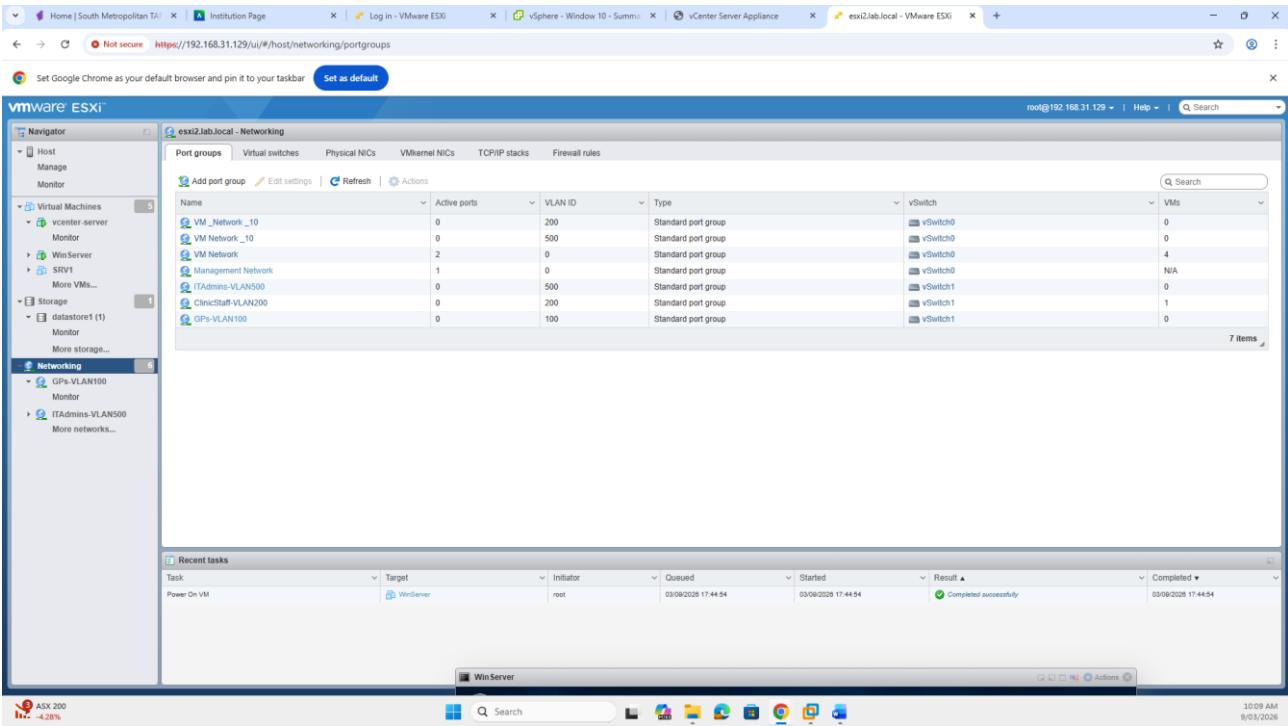


Figure 9 — VM Network port group configuration and accessibility status.

8. Connectivity Validation

The supplied evidence uses ping and DNS resolution tests to validate management and virtualisation connectivity. These screenshots are positioned here because they provide direct operational evidence rather than configuration-only evidence.

8.1 Host Connectivity Test

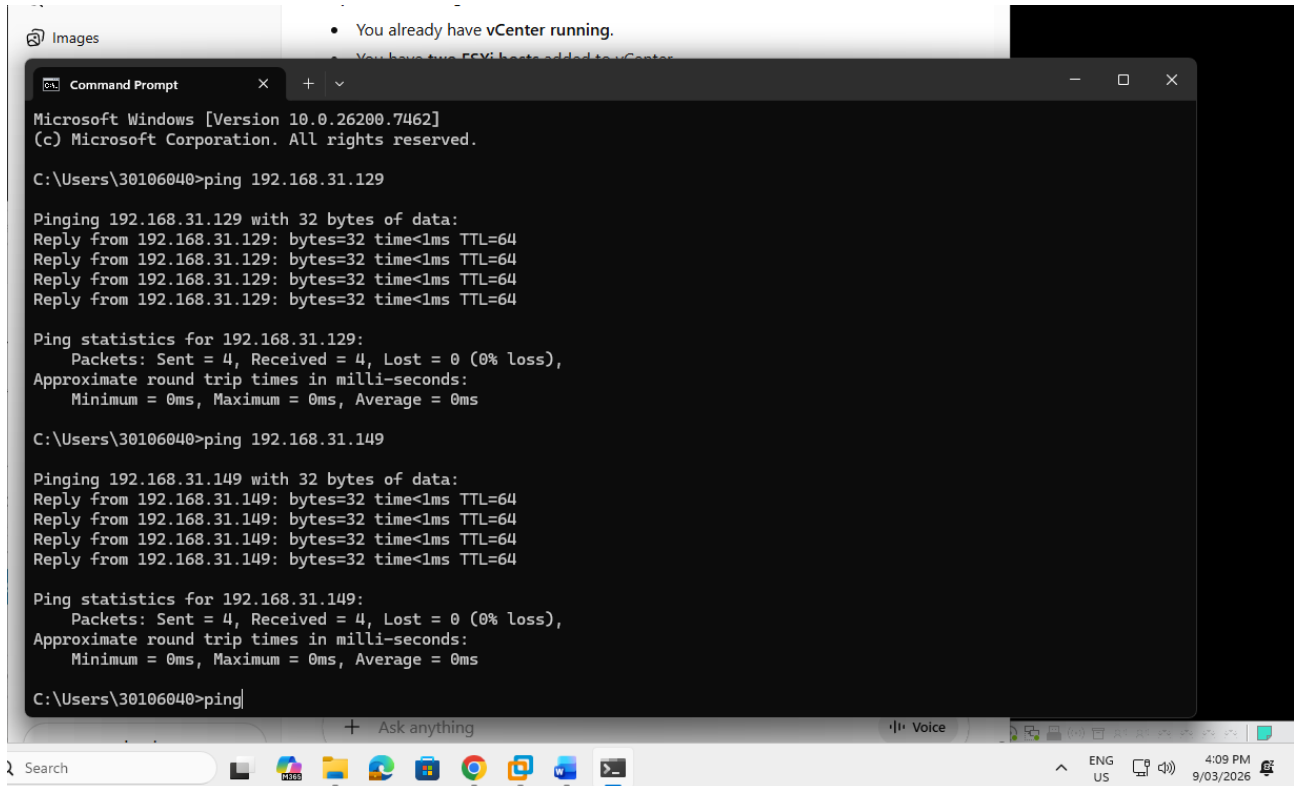
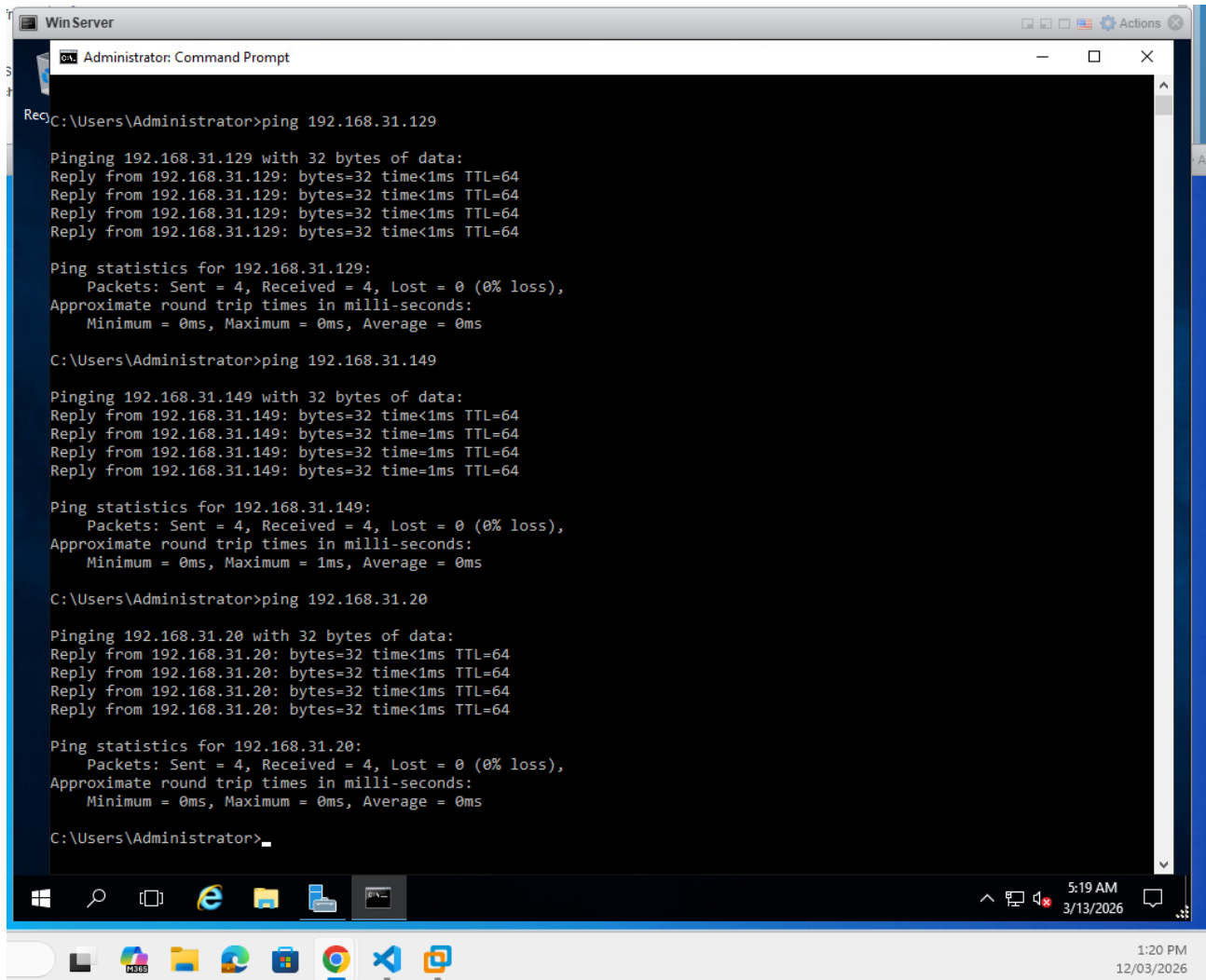


Figure 10 — Ping test to 192.168.31.129 with successful replies and 0% packet loss.

8.2 Additional Connectivity Validation



```
Win Server
Administrator: Command Prompt

C:\Users\Administrator>ping 192.168.31.129

Pinging 192.168.31.129 with 32 bytes of data:
Reply from 192.168.31.129: bytes=32 time<1ms TTL=64
Reply from 192.168.31.129: bytes=32 time<1ms TTL=64
Reply from 192.168.31.129: bytes=32 time<1ms TTL=64
Reply from 192.168.31.129: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.31.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>ping 192.168.31.149

Pinging 192.168.31.149 with 32 bytes of data:
Reply from 192.168.31.149: bytes=32 time<1ms TTL=64
Reply from 192.168.31.149: bytes=32 time<1ms TTL=64
Reply from 192.168.31.149: bytes=32 time<1ms TTL=64
Reply from 192.168.31.149: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.31.149:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Administrator>ping 192.168.31.20

Pinging 192.168.31.20 with 32 bytes of data:
Reply from 192.168.31.20: bytes=32 time<1ms TTL=64
Reply from 192.168.31.20: bytes=32 time<1ms TTL=64
Reply from 192.168.31.20: bytes=32 time<1ms TTL=64
Reply from 192.168.31.20: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.31.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>
```

Figure 11 — Windows command prompt connectivity test to 192.168.31.129.

8.3 DNS and Host Resolution

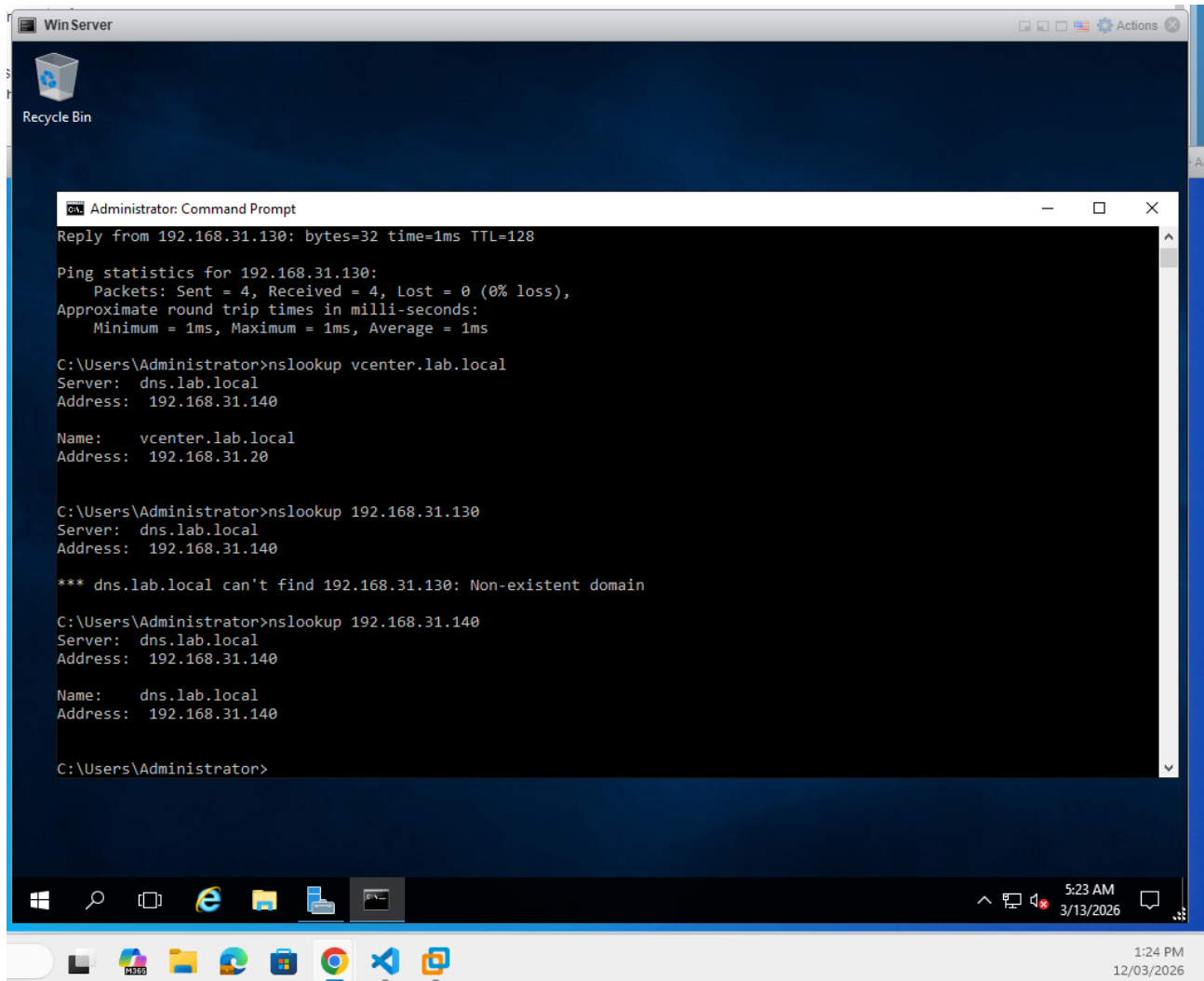


Figure 12 — Ping and nslookup evidence, including DNS server identification.

8.4 vCenter Name Resolution

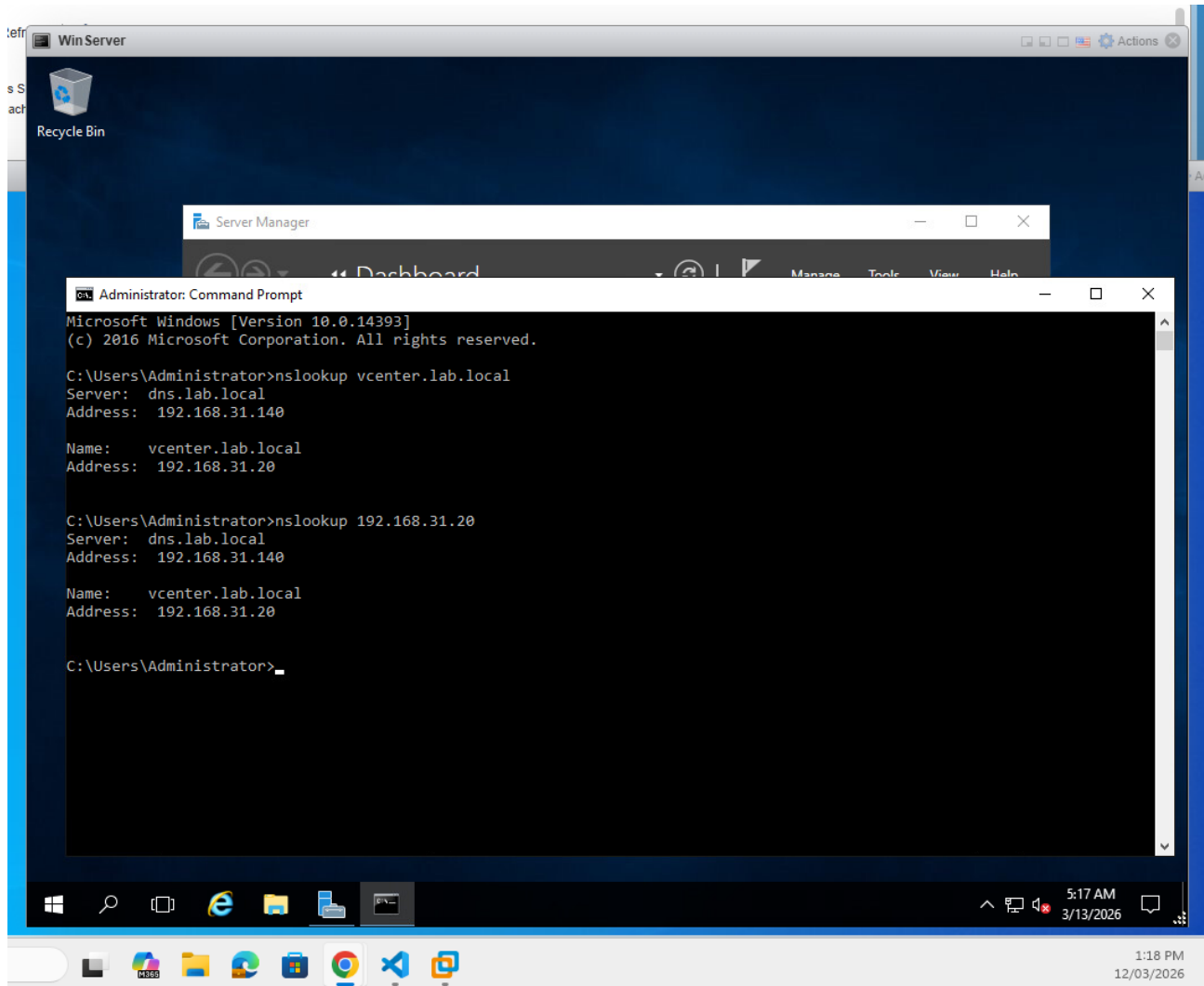


Figure 13 — nslookup resolving vcenter.lab.local to 192.168.31.20.

9. Shared NFS Storage

Shared storage is a core part of the project because it supports VM files, templates and installation media and enables host-level migration operations.

9.1 NFS Server Share

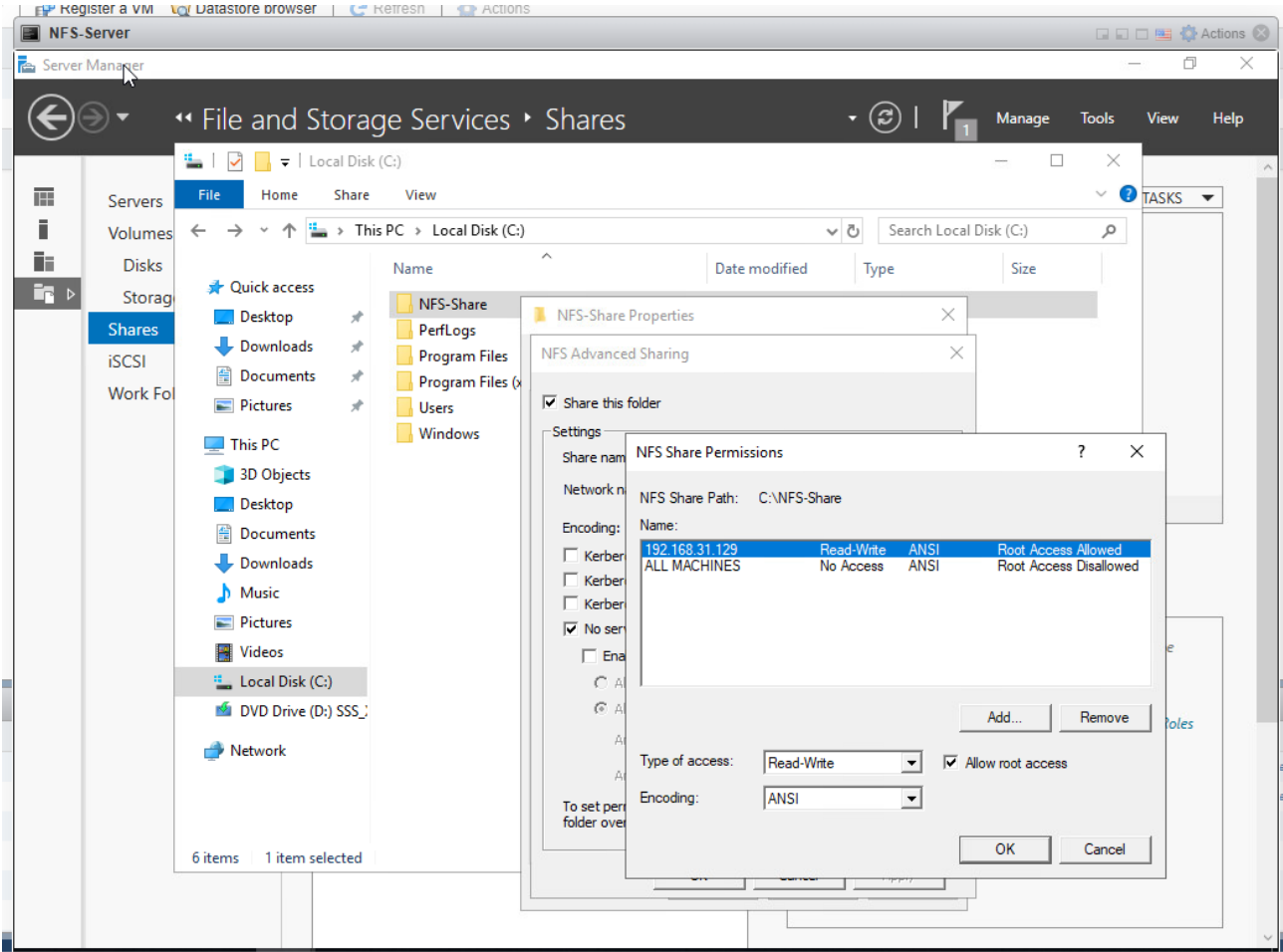


Figure 14 — NFS server storage/share configuration evidence.

9.2 NFS Share Contents

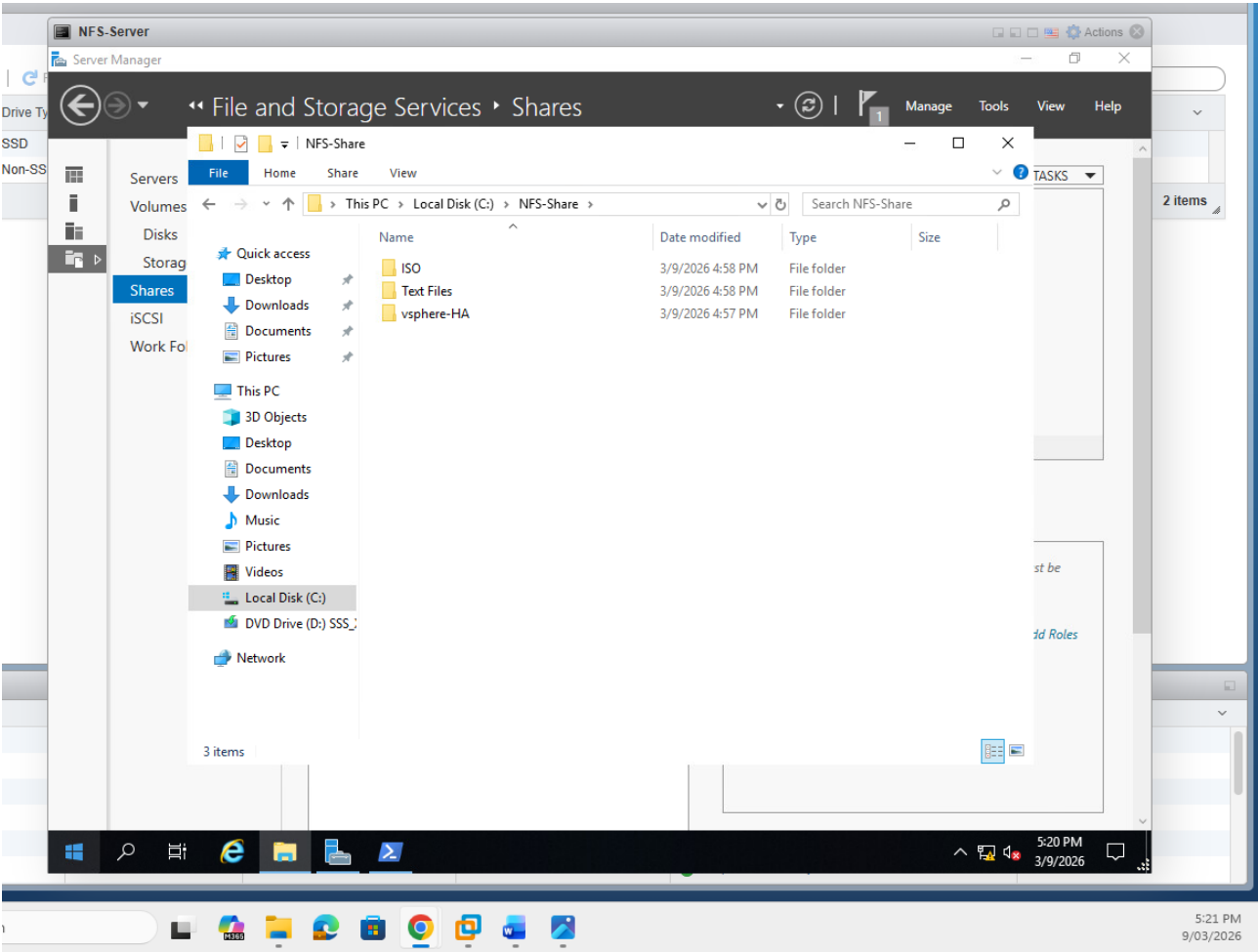


Figure 15 — NFS share contents used by the virtualisation environment.

10. Virtual Machine Provisioning

The provisioning workflow covers the creation of an ISO location, use of deployment resources and creation of virtual machines through the vSphere interface. The project brief requires templates/clones for Windows Server and Windows client VMs.

10.1 VM Inventory / Deployment Environment

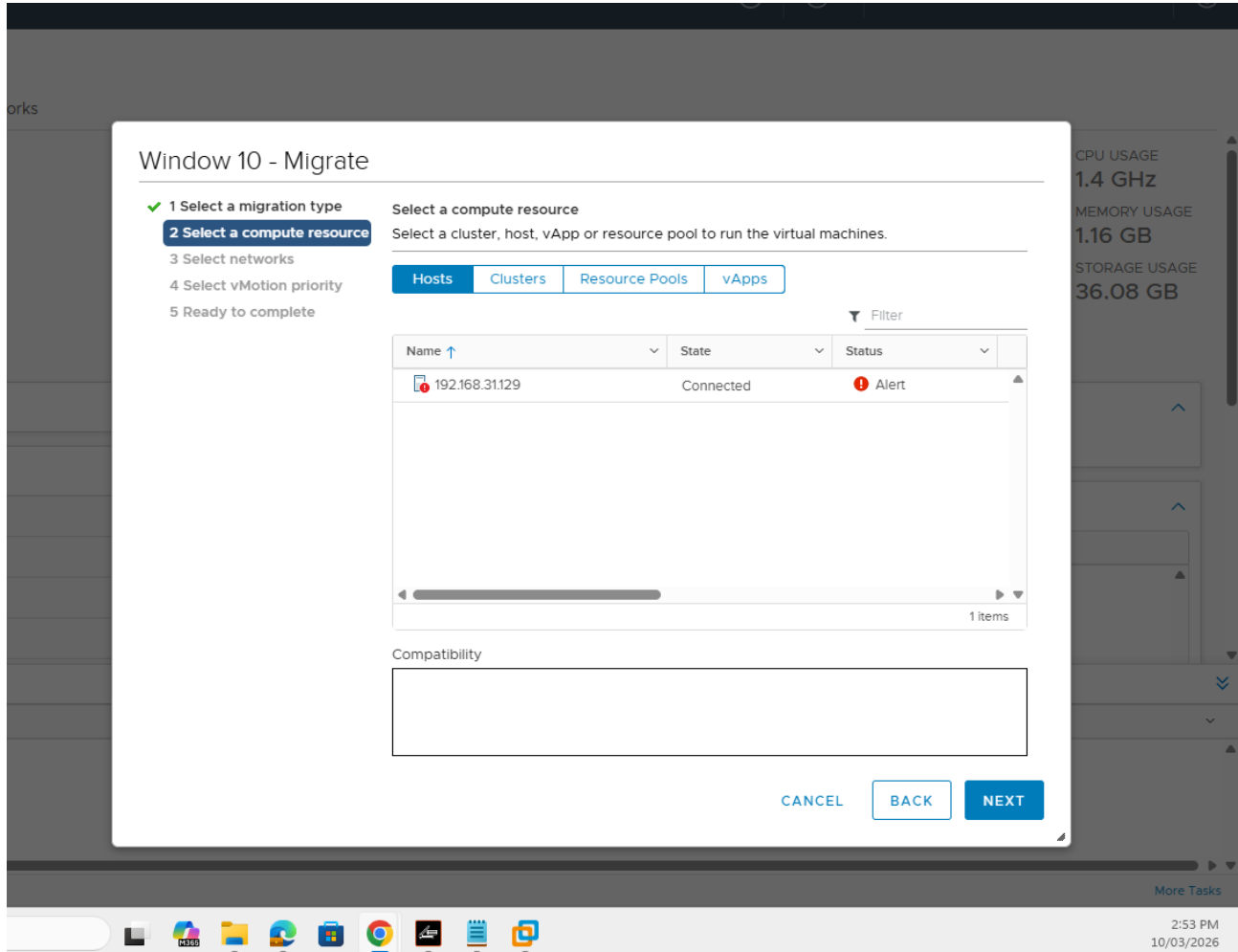


Figure 16 — ESXi virtual-machine inventory showing the deployed VM environment.

10.2 Migration / Provisioning Workflow

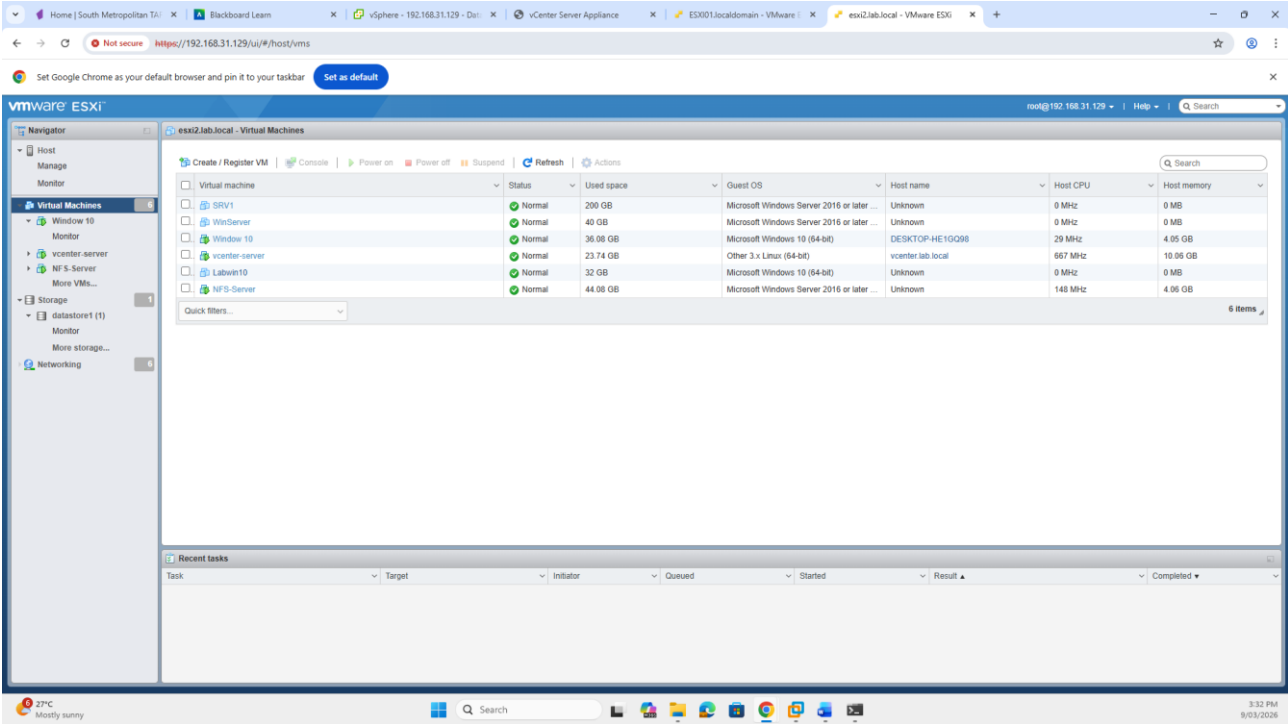


Figure 17 — vSphere migration wizard showing the migration workflow stages.

10.3 New Virtual Machine Workflow

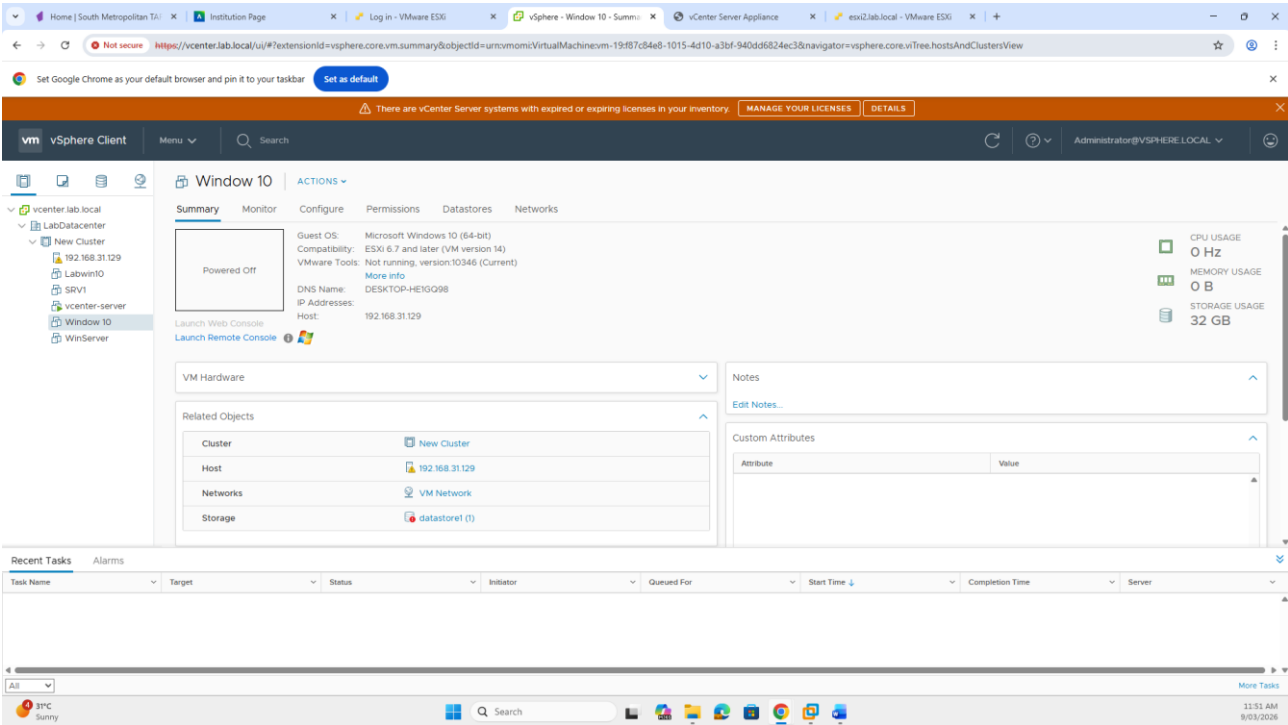


Figure 18 — New Virtual Machine workflow in vSphere Client.

10.4 Deployed VM Summary

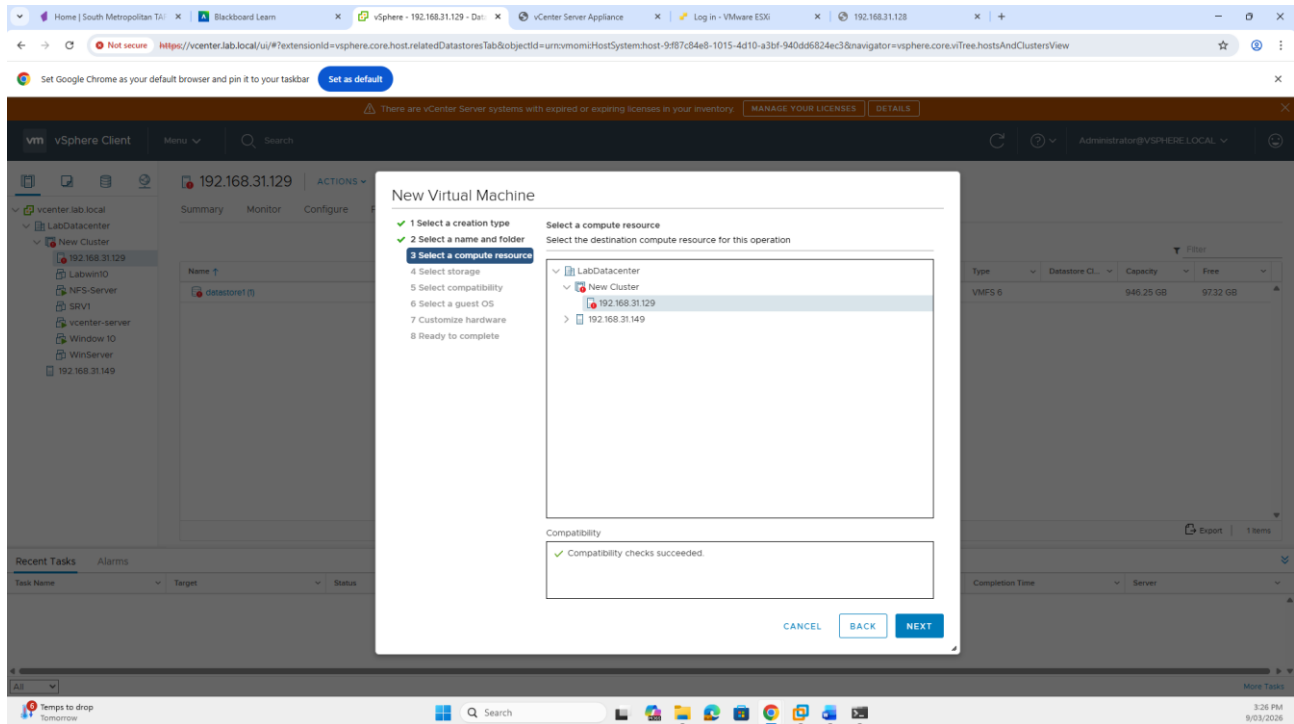


Figure 19 — vSphere Client summary for the deployed Windows 10 VM.

11. Migration, DRS and High Availability

The project requires cold and hot VM migration between multiple ESXi hosts and network storage, together with DRS and HA to manage resources and availability.

11.1 DRS Configuration

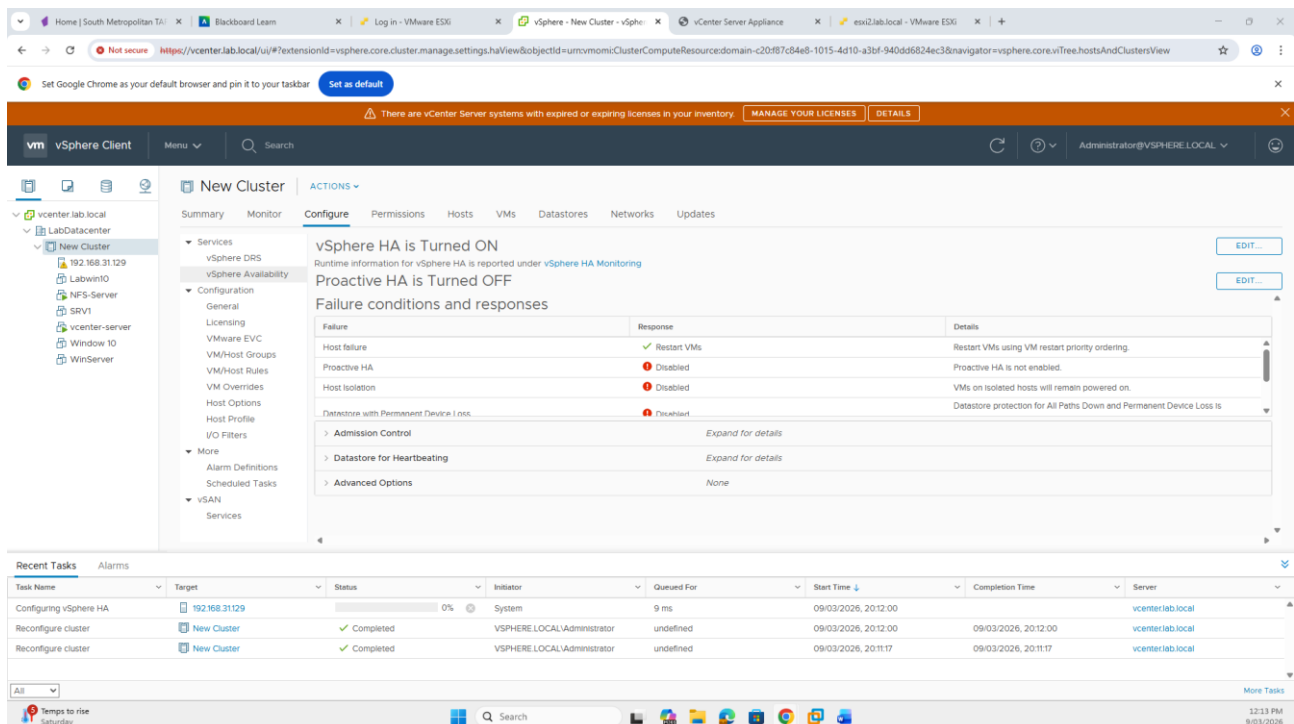


Figure 20 — vSphere DRS configuration for the cluster.

11.2 vSphere HA Configuration

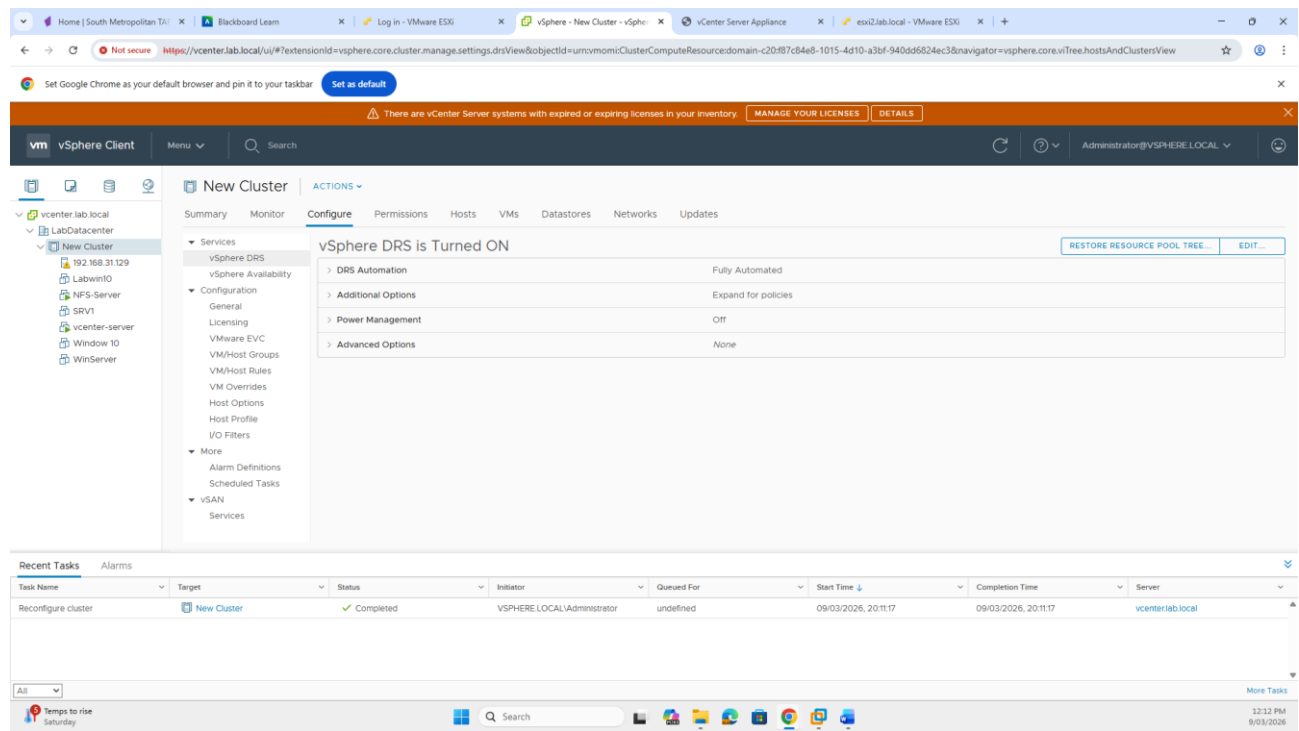


Figure 21 — vSphere HA configuration showing HA enabled and host-failure response settings.

Operational outcome: the supplied handover documentation records VM migration between hosts, automatic restart following host failure, resource load balancing and failover testing.

12. Final Handover Summary

The supplied handover documentation describes the completed environment as two physical ESXi hosts connected to managed network switches, with vCenter Server, a Domain Controller, an NFS/File Server and a Windows 10 client as core virtual machines.

Documented Outcomes

- Centralised management of servers.
- Reduced dependency on physical hardware.
- Improved disaster-recovery readiness.
- Enhanced security through segmentation and access control.
- Scalable design for future staff and system growth.

The original handover identifies hardware failure, storage limitations and configuration errors as key risks and records clustering/HA, capacity planning, documentation and testing as corresponding controls.

13. Evidence Placement Register

This register shows where each supplied screenshot is placed in the portfolio documentation.

Evidence	Screenshot	Documentation location
E01	Evidence 01	5.1 ESXi Host 2
E02	Evidence 02	5.2 ESXi Host 1
E03	Evidence 03	5.3 vCenter Server Appliance
E04	Evidence 04	6.2 Datastore Configuration
E05	Evidence 05	5.4 Windows Server / Domain Services
E06	Evidence 06	6.3 NFS Datastore Browser
E07	Evidence 07	7.1 Virtual Switches and Port Groups
E08	Evidence 08	7.2 VM Network / Port-Group Configuration
E09	Evidence 09	8.1 Host Connectivity Test
E10	Evidence 10	8.2 Additional Connectivity Validation
E11	Evidence 11	8.3 DNS and Host Resolution
E12	Evidence 12	8.4 vCenter Name Resolution
E13	Evidence 13	9.1 NFS Server Share
E14	Evidence 14	9.2 NFS Share Contents
E15	Evidence 15	10.1 VM Inventory / Deployment Environment
E16	Evidence 16	10.2 Migration / Provisioning Workflow
E17	Evidence 17	10.3 New Virtual Machine Workflow
E18	Evidence 18	10.4 Deployed VM Summary /

		vSphere Context
E19	Evidence 19	11.1 DRS Configuration
E20	Evidence 20	11.2 vSphere HA Configuration